

Econ 646 Spring 2011
Answer Keys to homework 7

- 1-7 Should be worked out by yourself.
 8. You already derived

$$p_t^* = \frac{1}{2}p_{t-1}^* + \frac{1}{2}E_t[p_{t+1}^*] + (\sigma + \varphi)(\tilde{y}_t + E_t[\tilde{y}_{t+1}])$$

Substitute the solution guess $p_t^* = \lambda p_{t-1}^* + (1 - \lambda)m_t$, the money demand function, the definition of p_t into the right hand side. After tedious calculation, you will find the relation

$$p_t^* = \frac{\frac{1}{2}(1 - \delta)}{1 + \delta - \frac{1}{2}(1 - \delta)\lambda} p_{t-1}^* + \left(1 - \frac{\frac{1}{2}(1 - \delta)}{1 + \delta - \frac{1}{2}(1 - \delta)\lambda}\right) m_t$$

where $\delta \equiv \sigma + \varphi$. For the initial guess to be consistent, λ has to satisfy

$$\lambda = \frac{\frac{1}{2}(1 - \delta)}{1 + \delta - \frac{1}{2}(1 - \delta)\lambda}$$

Solving this equation, you will find two roots:

$$\lambda = \frac{1 + \sqrt{\delta}}{1 - \sqrt{\delta}}$$

and

$$\lambda = \frac{1 - \sqrt{\delta}}{1 + \sqrt{\delta}}$$

The first root is apparently greater than one, and the second root is apparently smaller than one. Economically plausible one will be the second.

$$p_t^* = (1 - \lambda) \sum_{k=0}^{\infty} \lambda^k m_{t-k}$$

$$p_t = \frac{1}{2}(1 - \lambda) \left[m_t + \sum_{k=1}^{\infty} \lambda^{k-1} (1 + \lambda) m_{t-k} \right]$$

$$y_t = \frac{1}{2}(1 + \lambda) m_t - \frac{1}{2}(1 - \lambda) \sum_{k=1}^{\infty} \lambda^{k-1} (1 + \lambda) m_{t-k}$$

9. Yes. To make the argument simple, let's assume that δ is less than one (otherwise, λ is negative, implying that the model exhibits counter-intuitive oscillating impulse response functions). Now, let's suppose m_t increases by 1%. You can see from the expressions in 8 that general price level p_t increases in the same period, but the change is smaller than 1%. The money demand function $y_t = m_t - p_t$ implies that output increases.
10. Remember if there is no nominal rigidity, the classical dichotomy holds: real variables are determined by real factors. In our context, it means that 1% increase in m_t is accompanied by immediate 1% increase in p_t , and y_t does not change. That is the flexible price case.

Now think of the staggered price adjustment case. First, think about what is necessary for the aggregate price level to immediately increase by 1%. Because only half of the firms has price changing opportunities, these firms have to adjust their prices by 2%. But that is not what they want, because if they do so, their prices becomes ridiculously high compared to those who do not change prices, losing lots of demands.

Then, how much do price changing firms adjust their prices. It turns out that, from the expression in 8, they change their prices *less than 1%*. This is because what they care about is relative prices. To picture it, think of two firms, you and your neighbor. If you know your neighbor does not change her price today, you do not want to change your price very aggressively, because you care about your price relative to your neighbor's.

11. Yes. Again, that is apparent from the expression in 8. 1% increase in m_t does not increase the general price level p_t by 1% in one period later.

12. I already argued that, if you know your neighbor does not change her price in the period of nominal shock, you change your price less than 1%. One period later, you and your neighbor switch roles: it's your neighbor's turn. How much does she adjust her price? It turns out that, again from the expression in 8, she changes her price *less than 1%*. This is because she cares about her price relative to yours. The fact that you changed your price so little in the previous period then reduces her incentive to adjust her price now. The aggregate price level does not increase by 1%, so output increases.

The argument continues. In the next period, you have your second price changing opportunity. You find your neighbor's price is still low, so you don't want to change your price very much. Next, your neighbor. She finds your price is still low, changing her price little. The argument goes on. Actually, the aggregate price level increases by 1% only in the limit.